MIL-G-51478A(EA)
4 June 1987
SUPERSEDING
MIL-G-51478(EA)
18 February 1983

MILITARY SPECIFICATION

GRENADE, LAUNCHER, SMOKE: SCREENING, RP, UK L8A3

This specification is approved for use within the Chemical Research, Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of grenade.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1234	-	Pyrotechnics: Sampling, Inspection and Testing

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 1330

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2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING AND TECHNICAL DATA PACKAGE LIST (TDPL)

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

TDPL 13-19-100 - Grenade, Launcher, Smoke: Screening, RP, UK L8A3

(Copies of standards and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 13-19-100 or on the subsidiary drawings shall conform to the specification listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the grenade shall conform to the specifications and drawings listed on TDPL 13-19-100 and on subsidiary drawings.

3.2 Assembly. The grenade shall be assembled as shown on Drawing 13-19-100.

3.2.1 Subassemblies.

3.2.1.1 Delay assembly. All components of the delay assembly and cylinder shall be assembled as shown on Drawing 13-19-102. The delay shall burn for 0.95 to 1.01 seconds when tested in accordance with 4.4.4.1. The delay burn time, together with the weight of the black powder propulsion charge, controls the grenade burst and height distances.

3.2.1.2 Cylinder assembly. All components of the cylinder assembly shall be assembled as shown on Drawing 13-19-101. The cylinder assembly shall have an outer clip to inner clip resistance of 0.9 to 1.6 ohms when tested in accordance with 4.4.4.2.

3.2.1.3 Moisture and volatiles. When tested as specified in 4.4.4.5, the moisture and volatiles content of the red phosphorus butyl rubber shall be no greater than 0.15 percent by weight and that of the black powder shall be no greater than 0.70 percent by weight when loaded into the grenade.

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3.2.1.4 Red phosphorus butyl rubber characteristics. The red phosphorus butyl rubber shall conform to drawing requirements when tested as specified in 4.4.4.6.

3.3 Immersion. The grenade shall not have evidence of leakage after water immersion and shall conform to the requirements of 3.4 when tested in accordance with 4.4.4.3.

3.4 Functioning. The grenade shall function as follows when tested in accordance with 4.4.4.3 and 4.4.4.4.

(a) The electric fuze shall function igniting the propellant charge and the delay composition. The gas from the burning propellant shall displace or rupture the diaphragm and launch the grenade from the discharger. During the flight of the grenade, the delay composition shall burn through and ignite the burster charge. The grenade shall burst 23 meters to 33 meters from the discharger at a height of 6 meters to 12 meters.

(b) When the grenade bursts, the burst shall break the red phosphorus/butyl rubber pellet into pieces approximately the size of the original granules and ignite these granular pieces. The burst shall then rupture the rubber body and scatter the burning pieces. No more than 25 grams of unignited granules shall remain in the grenade's rubber body. The granules shall not scatter more than 75 meters from the discharger. The burning, falling granules shall reinforce the smoke cloud by continuing to burn for no more than 6 minutes.

(c) The time from grenade launch to good obscuration, as a result of the grenade's smoke, shall be no more than 6 seconds. The smoke shall provide good obscuration for not less than 60 seconds as assessed by an experienced observer.

3.5 Preproduction. Prior to the start of regular production, a preproduction lot of grenades shall be produced, examined and tested in accordance with this specification (see 4.3).

3.6 Workmanship. The grenade shall be free from smoke mix contamination, foreign matter and damage such as dents, cracks or deformation which could impair its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 600 grenades or production from one shift, whichever is greater, shall be produced using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105, level III.

4.3.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

4.3.2.3 For black powder and red phosphorus butyl rubber tests. Each batch of red phosphorus butyl rubber and black powder used in the preproduction lot of grenades shall be sampled.

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive test. Sample grenades shall be examined in accordance with the classification of defects and MIL-STD-105.

4.3.3.2 For destructive tests. Sample grenades shall be tested in accordance with table I.

4.3.3.3 For black powder and red phosphorus butyl rubber tests. The black powder sample shall be tested as specified in 4.4.4.5. The red phosphorus butyl rubber sample shall be tested as specified in 4.4.4.5 and 4.4.4.6.

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Failure to pass tests shall be cause for rejection of the batch of material represented.

TABLE I. Destructive tests

Test	Requirement	Method	Accept	Reject
Grenade launch	3.4	4.4.4.4	0	1
Grenade burst	3.4	4.4.4.4	0	1
Grenade burst distance 23 to 33 meters	3.4(a)	4.4.4.4	1	2
Grenade burst height 6 to 12 meters	3.4(a)	4.4.4.4	1	2
Unignited red phosphorus in grenade body 25 grams maximum	3.4(b)	4.4.4.4	2	3
Red phosphorus particle range 75 meters maximum	3.4(b)	4.4.4.4	2	3
Red phosphorus burn time 6 minutes maximum	3.4(b)	4.4.4.4	2	3
Time from launch to good obscuration 6 seconds maximum	3.4(c)	4.4.4.4	1	2
Duration of good obscuration 60 seconds minimum	3.4(c)	4.4.4.4	0	1

NOTE: Select samples in accordance with MIL-STD-105, level S-4, AQL 1.5 percent defective.

4.3.4 Rejection criteria. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the preproduction lot shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot of grenades shall be identified and controlled in accordance with MIL-STD-1168, produced by one manufacturer in no more than five

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consecutive work shifts not to exceed 2 weeks consecutive duration from the same materials and under essentially the same manufacturing conditions. However, no more than one lot interfix number of metal, rubber, or plastic parts, no more than one lot of fuzes or black powder, and no more than two lots of smoke mix ingredients from one manufacturer shall be represented in any one lot of assembled grenades. The minimum lot size shall be 1,201 grenades.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive test. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

4.4.2.3 For black powder test. A sample shall be selected from the black powder used for each day's production.

4.4.2.4 For red phosphorus butyl rubber tests. One batch of red phosphorus butyl rubber from each day's production or one batch from each ten batches produced, whichever comes first, shall be sampled. A nonconforming batch shall cause each batch produced since the last batch that was sampled to be sampled and tested.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample grenades and the packing thereof shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For destructive test. Sample grenades shall be tested in accordance with table I. Delay burn characteristic shall be tested in accordance with 4.4.4.1. Delay burn time shall meet an AQL of 0.40 percent defective and shall be accepted prior to assembly into grenades. Note: This delay test may be accomplished using continuous sampling techniques, if appropriate.

4.4.3.3 For black powder test. The black powder sample shall be tested as specified in 4.4.4.5. If moisture and volatiles exceed requirements, dry and retest prior to use.

4.4.3.4 For red phosphorus butyl rubber tests. The red phosphorus butyl rubber sample shall be tested as specified in 4.4.4.5 and 4.4.4.6. If moisture and volatiles exceed requirements, dry and retest prior to use. Failure to conform to any other drawing requirements shall be cause for rejection of the batch represented.

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4.4.3.5 Classification of defects.

(a) Grenade, launcher, smoke: screening, RP, UK L8A3 (Drawing 13-19-100).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Marking missing or incorrect	
103	Diaphragm incorrect	
104	Cylinder assembly resistance incorrect	4.4.4.2
105	Workmanship (3.6)	
106	Clip damaged	

(b) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Unit container closure incorrect
102	Marking missing, incorrect or illegible
103	Shipping container closure incorrect
104	Shipping container loosely packed
105	Strapping missing or loose

4.4.4 Tests.

4.4.4.1 Delay assembly. The delay shall be ignited and the delay time measured. The test equipment shall be capable of measuring the burning time of the delay to the nearest 0.01 second.

4.4.4.2 Cylinder assembly. The electrical resistance of the contact assembly shall be measured using an ohmmeter with a maximum short circuit current of less than 10 milliamperes (ma). Tests shall be conducted in a chamber strong enough to contain any accidental combustion products, should ignition occur. Proper safety precautions shall be incorporated to protect personnel during all phases of this test. Care shall be taken to avoid damaging or bending the electrical contact spring clips.

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4.4.4.3 Immersion. The grenade shall be heated for a minimum period of 8 hours at a temperature of 45° ± 5°C and immersed in 0.9 meter minimum of water at 18° ± 5°C for a minimum period of 2 hours. Within 1 hour subsequent to removal from water, the grenade shall be functioned.

4.4.4.4 Functioning. The grenade shall be functioned from a function Tester, Grenade Ignition and Propulsion, Q226, Drawing 136-33-1, or from an Army discharger, Drawing 13-12-34, 13-12-35, or 13-12-90, or from a similar device designed by the contractor and approved by the Government.

(a) The muzzle of the launcher shall be 2 to 2.3 meters above ground level at an elevation of 23° ± 1/2°.
(b) The terrain shall be soft, dry, level and fire resistant.
(c) Wind speed shall not exceed 24 kilometers per hour (km/h).
(d) Grenade shall be initiated by a DC electrical source with an average potential of 24 volts.

4.4.4.5 Moisture and volatiles. The moisture and volatiles determination shall be conducted in accordance with method 101.1, 101.2, or 102.2 of MIL-STD-1234.

4.4.4.6 Red phosphorus butyl rubber characteristics. The red phosphorus butyl rubber shall be analyzed for conformance to drawing requirements after obtaining Government approval of the test methods to be used.

5. PACKAGING

5.1 Packing, level A. Four grenades shall be packed together as shown on Drawing 13-19-128.

5.2 Marking. Each ammunition container shall be marked as shown on Drawing 13-19-128.

5.3 Palletization. Ninety-six packed ammunition containers (384 grenades) shall be palletized as shown on Drawing 13-19-132.

6. NOTES

6.1 Intended use. This grenade is intended to be used to produce a visual smoke screen.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Preproduction information:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract
(2) Name and address of test facility and shipping instructions when testing is performed by the Government

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(3) Time required for the Government to notify the contractor whether to proceed with production

6.3 Precautions. This specification covers inspection of explosive and chemical materials (black powder, low explosive; red phosphorus/butyl rubber granules, FLAMMABLE SOLID) which are potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of these materials.

6.4 Subject term (key word listing).

Black powder
Butyl rubber
Red phosphorus
Smoke grenade

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1330-A316

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-51478A(EA)	GRENADE, LAUNCHER, SMOKE: SCREENING, RP, UK L8A3

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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